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the others so that they, too, can have a better solution to their research problem—just like the efforts of countless individuals who developed ns-3 helped us.

I would now like to mention a few topics that didn't make it into this series of articles on ns-3. The one that comes to my mind first is PyViz, a live simulation visualiser. It doesn't require any trace files to generate the animation. PyViz is mostly written in Python, but works with both Python scripts and C++ scripts. It is mostly used for debugging purposes. With it, we can check the working of mobility models and the time at which packets are dropped. PyViz will not work without Python bindings. So we have one point in favour of Python bindings.

Another area I skipped altogether is the installation of ns-3 using Cygwin (a utility which helps Linux applications to run on Windows). Over the years, I have faced the following question many times. Is it possible to install ns-2 and ns-3 on Microsoft Windows operating systems? Yes. It is possible to do so. But the question is, why? Cygwin based ns-3 installation is a tedious task. There might be compatibility issues between POSIX APIs and Win32 system calls. You might come across unexpected errors frequently. People hope that by installing ns-3 on Windows, the whole process of network simulation will become easier. This is a false hope; whether the installation is on Linux or Windows makes no difference to ns-3.

Another area I haven't covered is third party contributions to ns-3. Many research groups have contributed to the code base of ns-3. For example, cognitive radio extension for ns-3 called CRE-NS3 is maintained as contributed code by third party developers. There might be other interesting topics from ns-3 left out in this series for want of space. Out there is an ocean and I have just provided a cupful, so do forgive me.

You can find many useful ns-3 documents at www.nsnam.org/documentation/. Before concluding, a word about the way this series was organised. I haven't explained the finer details of ns-3; for that you have a tutorial available in the above mentioned location. I haven't elaborated much about the complex structure of ns-3; for that you have the manual available in the same location. My intention was to present the case for ns-3, guide every one through the installation process, familiarise every one with the different tools associated with ns-3 and, finally, warn you about a few traps and pitfalls in ns-3—most of which I myself have fallen prey to. **END** 

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